

Darren Liu

Biophysics Ph.D. Graduate Student | University of Chicago | Chicago, IL 60637

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Summary

- Interdisciplinary scientist with practical experience in AI/ML and wet lab biology, skilled at bridging experimental and computational teams to drive cross-functional collaboration and innovation.
- Molecular biologist with expertise in high-throughput assay development, automated liquid handling, and molecular evolution.
- Biophysicist skilled in developing predictive models for cell culture development, utilizing protein language models, and building biologically informed models of protein-protein interactions.

Education

University of Chicago

2022-Present

Ph.D. Biophysical Sciences. PI: Aaron Dinner and Arvind Murugan

- Research Interests: protein-protein interactions, protein evolution, automated liquid handling, dimensionality reduction, protein language models, chemical biology, machine learning, mechanobiology, molecular dynamics, protein structure modeling

University of Chicago

2022-2023

M.S. Biophysical Sciences

Boston University

2018-2022

B.A. in Chemistry: Biochemistry. Double minors in Biology and Computer Science

Research and Work Experience

AI/ML CMC Cell Culture Development Intern @ AbbVie

Summer of 2026

Manager: Anh Nguyen Dang. CMC Biologics Drug Substance Intern - Cell Culture Development

- Developed a hybrid ML/ODE model to predict CHO cell culture time series data (VCD, titer, glucose, lactate) to accelerate FIH timelines
- Trained models to leverage protein language model sequence embeddings and biophysical properties of therapeutic antibodies to enable cross-molecule prediction of cell culture process data.
- Aligned modeling via cross-functional collaboration with experimentalists in cell culture and purification.

Graduate Researcher @ University of Chicago

2022-Present

PI: Aaron Dinner, Arvind Murugan, Department of Biophysical Sciences

- Developed a high-throughput plaque assay workflow integrating viral molecular evolution, liquid handling, and image segmentation to assay how protein-protein interactions change with mutation.
- Built several families of Ising models to identify key parameters governing protein evolvability
- Analyzed biophysical data and protein-protein docking results to uncover the molecular mechanism of mechanosensitivity in actin binding proteins.

Undergraduate Researcher @ Boston University School of Medicine

2022

PI: William Lehman, Department of Pharmacology, Physiology & Biophysics

- Simulated tropomyosin point mutations with molecular dynamics (NAMD)
- Connected molecular changes to disease phenotypes (hypertrophic cardiomyopathy)

Undergraduate Researcher @ Stanford University

Summers of 2018-2022

PI: Julia Salzman, Department of Biochemistry

- Implemented classical and machine learning image segmentation methods for spatial transcriptomics.
- Mapped relationships between gene expression and viral load in COVID-positive patients
- Developed and evaluated models to extract RNA isoform information from fluorescence imaging data (MERFISH)

Teaching and Service

UChicago Computational Biology Outreach Volunteer

2022-Present

Taught computational biology workshops in Python and genetics to students ranging from elementary through high school, covering topics from the fundamentals of genome-wide association studies (GWAS) to phylogenetic trees.

Center for Living Systems Chalk Talk Organizer

2025-Present

Organizer for Chalk Talk seminar series

Biophysical Sciences Student Advisory Board Member

2024-Present

Elected by peers to coordinate and lead program-wide graduate student events, organizing recruitment events and community building activities.

Expanding Your Horizons (EYH) Volunteer

2022-Present

Supported STEM outreach by mentoring schoolgirls in hands-on workshops.

University of Chicago Teaching Assistant

2024

Created assignments, graded coursework, and lectured for "Computational Biology in Microbial Ecosystems" with PI Seppe Kuehn

@rtifice

2022-2023

Volunteer. Taught middle school kids basics of circuits and coding

GirlsWhoCode

2022

Volunteer. Taught fundamentals of coding to girls in elementary and middle school

Boston University Chemia

2020-2022

Volunteer. Tutored undergraduates in organic and physical chemistry

MVLA Aspire

2017

Volunteer. Provided free ACT tutoring for students

Skills

- **Programming:** Python (NumPy, PyTorch, SciPy, Biopython, Jupyter, Sci-kit Learn), High-Performance Computing (SLURM, bash), Git, LaTeX
- **Wet Lab:** Assay development, automated liquid handling (Tecan Fluent, Echo Acoustic Liquid Handler), plate readers (fluorescence and absorbance assays), molecular biology, PCR, DNA assemblies, DNA cloning, DNA sequencing, directed molecular evolution, cell culture, chemical biology
- **Computation:** Dimensionality reduction, clustering, regression, protein language models (ESM2), gaussian process, AlphaFold2, ClusPro, ordinary differential equations (ODE), molecular dynamics (NAMD), VMD/PyMOL, image segmentation (ImageJ, OpenCV, scikit-image, PyTorch), sequencing analysis

Publications

Halder, S.S., Bellitto, J.G., Rynkiewicz, M.J., Cheung, F., **Liu, D.**, Firlar, I., Bongiorno, A., Barry, M.E., Sewanan, L.R., Moore, J.R., et al. (2025). Evaluating pathogenicity of TPM1 variants of unknown significance using in-silico and in-vitro models. *The Journal of Precision Medicine: Health and Disease* 2, 100008

Presentations and Posters

Mapping the evolvability of molecular specificity across a protein family. **Poster, Gordon Research Conference on Molecular Mechanisms in Evolution** (June 2025)